

test drive ■ motherboards

The mid-range category for the AMD platform comprised five boards from four vendors. Out of these, three were based on nVidia's nForce2 chipset, and the other two were based on VIA's relatively new KT600 and the older KT400 chipset. Prices began from as low as Rs 5,700, depending on features and package contents, moved upwards. We also received a limited edition board from Gigabyte, priced at Rs 13,000, which, according to our categorisation based on price, should fit in the high-end category, but taking into account its features and performance, we compared it with boards in the mid-range category.

The board based on VIA's new KT600 was the first board based on this chipset we reviewed. We were eager to see this updated chipset take on nVidia's powerhouse chipset. Besides, the board came from ASUS, adding to our expectation of it emerging the winner. But in the end, the fight was between nVidia's nForce2 and VIA's KT600.

Features

Boards based on the nForce2 were rich in features, and comparable to Intel's mid-range section. They support a CPU running at 200 MHz FSB, which means you can plug in an Athlon XP 3200+, currently the fastest available 32-bit processor from AMD. All these boards have support for DDR400 memory, and they run in dual-channel mode, thus spelling higher bandwidth for memory-intensive applications. With the exception of the Gigabyte GA-7NNXP Limited Edition board—which has four memory slots—all the others had three slots. The ASUS A7V600 supports a 200 MHz FSB CPU and DDR 400 memory, and the AOpen AK77-8X, based on the KT400, supports a CPU at 166 MHz FSB—

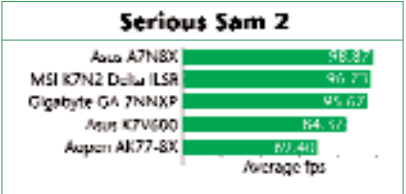
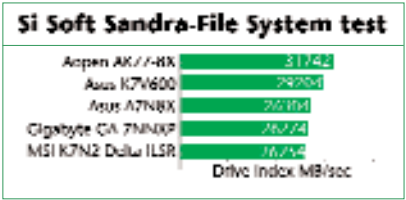


The Gigabyte GA-7NNXP is a special edition board that's literally bursting with all the features you'd ever need

for example, the Athlon XP 2800+—and DDR 333 MHz memory. Unofficially, though, it supports DDR400.

All the boards were based on the ATX form factor, and only MSI did a great job of the available real estate. The CPU area didn't have any obstructive capacitors, so you can plug on a bigger heat sink-fan combo with ease. The same went for the ASUS A7N8X. The extra dual-power supply daughter card slot on the Gigabyte board made it a little difficult for clipping on a bigger heat sink. The VIA-based boards were average in their design approach.

The boards from ASUS, Gigabyte and MSI, based on the nForce2, have the feature-rich MCP-T Southbridge, so there were Firewire ports and the audio processing unit on these boards. There was, however, no sign of the newer Firewire 1394B standard. The AOpen and ASUS boards, based on VIA chipsets, lacked these features entirely. The Ultra versions of the nForce2 chipset support dual LAN; only ASUS' A7N8X and Gigabyte's 8KNNXP boards were equipped with it, and offered both, Gigabit and Ethernet connectors. The MSI



Memo...Rising

The GHz value of a processor has always been used as a yardstick to determine a PC's performance. Similarly, the speed of system memory is another parameter that can be used to define computing power. These two parameters are inter-linked, and hence depend upon each other to achieve the optimum computing speed. System memory, or RAM, has come a long way from the days of the slower EDO type, to today's blazing fast DDR standard and now the Dual channel implementation. But why did the shift happen in the first place?

Not so long ago, when processor speeds were still in the MHz, SDRAM—the de-facto standard then—ran at 133 MHz. But as processor speeds crossed 1 GHz, the RAM couldn't keep up with the processor's ability to process data and hence became a bottleneck for system performance. The problem was checked with the arrival of the DDR standard, which currently offers a speed of 400 MHz and above. Even such speeds do not offer the ideal throughput between memory and CPU. So, Intel decided to bring in dual-channel memory—extensively used in servers and workstation systems, to desktop systems.

To understand the difference between dual-channel and single channel memory, think of dual-channel memory as a two lane road, while single channel memory is a single lane road. The traffic (data)

on the two-lane road is handled more efficiently than on a single lane road, since it offers more width and two separate parallel paths for seamless motion. Similarly, dual-channel memory offers a wider data path (128-bit as against 64-bit for single channel memory), and two clear channels. Like traffic signals at junctions, the memory controller on the Northbridge (chipset) acts as a regulator, controlling the data transfer between CPU and memory. A single channel can be made to address a pair of memory modules, thus with two channels, you have four memory modules.

Memory performance is given in terms of Peak Data Bandwidth, which is the measure of throughput that the memory can handle. It can be calculated as:

Memory speed x Number of bytes transferred per channel x Number of channels.
 For PC3200 (DDR 400 in dual-channel implementation) RAM
 Peak Bandwidth = 400 MHz x 8 Bytes x 2 channels
 = 6,400 MBps
 = 6.4 GBps as compared to 3.2 GBps for a single channel implementation of the same memory.

Thus, dual channel memory doubles the maximum amount of data that can be transferred between the CPU and the system memory modules.